

ABSTRACT

Note: This project was developed using Android libraries from 2023. Due to recent API and library version changes, the project is currently not runnable without refactoring. The core goal was to implement on-device object detection using YOLOv8 and build an interactive translation interface. The full source code and UI screenshots are available in this repository.

This thesis presents the design, development, and implementation of Gözlük, an innovative Android application that integrates advanced machine learning techniques to provide a comprehensive translation service. The application is designed to overcome language barriers and improve communication by translating text into multiple languages, recognizing objects from images, and providing text-to-speech capabilities. Gözlük distinguishes itself from traditional translation apps by offering a unique feature: the ability to recognize objects in images and translate them. This feature can be particularly useful in various scenarios, such as educational environments, assisting visually impaired users, or helping travelers navigate unfamiliar environments. The app also features speech-to-text technology, enabling text-to-speech translation. This feature allows users to hear translations audibly, making the app more accessible and user-friendly, especially for visually impaired users or those who prefer auditory learning. A significant contribution of this project is the successful training and implementation of the YOLOv8 model, a state-of-the-art object detection algorithm. The trained model was then converted to TensorFlow Lite, a format optimized for mobile devices. This conversion enables the app to run the model efficiently, providing real-time object detection and translation on Android devices. As a result, Gözlük serves as a testament to the potential of integrating machine learning with mobile applications. The application serves as a practical example of how advanced machine learning techniques can be leveraged to create innovative solutions that address real-world challenges.



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Tespit Edilen Nesne :

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